

Class 5: Integrated Sensing and Communication

What it is, why it is hard, and how OFDM does it

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CIS 4930/5930 Future Edge Networks

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Two machines, one sky



Frankfurt airport surveillance radar

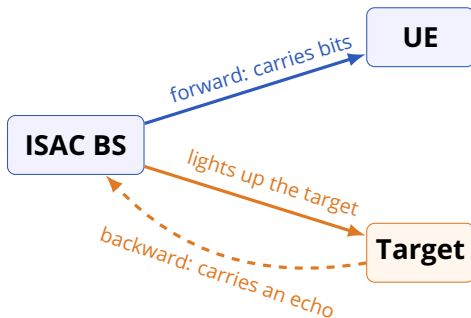


Ericsson active 5G antennas

Q: One sends a wave to be understood. One sends a wave to come back. What do they share?

A: Spectrum, power, antennas, and silicon — all of it scarce. Building two systems that never talk to each other pays for the same hardware twice.

One waveform, two jobs



The same transmission is **information** on the way out
and **measurement** on the way back.

ISAC does not add a radar to the network.
It reads the network's own signal as radar.

An echo is data nobody uploaded



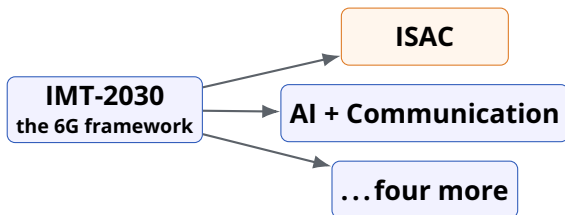
Waymo Jaguar I-Pace roof sensor package

- **Vehicles:** see what the camera cannot.
- **XR:** track a hand, no wearable.
- **IoT:** count people, no camera.
- **Space:** range the satellite you talk to.
- **Safety:** a drone, a fall, a flood.
- **Farming, weather, health:** same echo.

Every one of these is the network noticing something on its own.

XR = Extended Reality IoT = Internet of Things

Sensing is written into the 6G frame



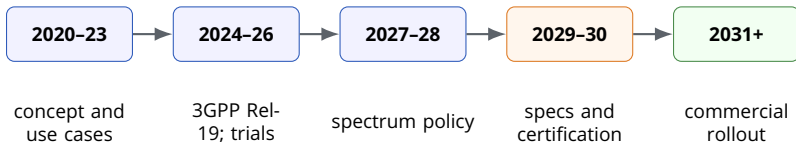
Q: Why does a standards body have to say this out loud?

A: Because a usage scenario is what gets spectrum, performance requirements, and test procedures written for it. Until ISAC was named, no radio was obliged to be measurable.

ITU-R Recommendation M.2160, IMT-2030 framework (2023)

Being named starts a clock

Being written into M.2160 does not ship a product; it starts a schedule — spectrum, requirements, tests, then certification. Here is where that schedule stood when this material was prepared.

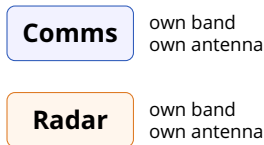


The 2020–2026 columns describe work that has happened. Everything from 2027 rightward is a forecast made in this talk, not a published schedule.

Standardisation is where the tradeoff stops being a curve and starts being a number somebody must certify.

Two boxes became one box

Separate systems



DFRC



What it buys

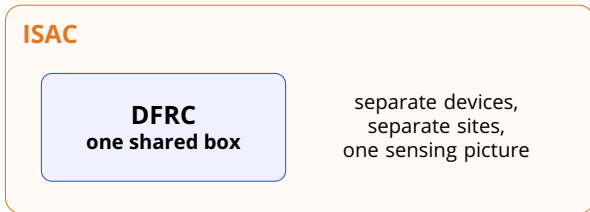
One spectrum licence, one amplifier, one aperture, one power budget.

DFRC = Dual-Function Radar Communication

What it costs

A single waveform now has to satisfy two specifications at once.

DFRC is one box. ISAC is the question.

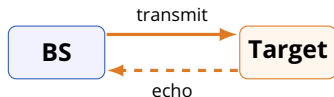


DFRC shares waveforms, antennas, and processing chains inside one radio.

ISAC also lets one node transmit and a different node listen.

Every DFRC system is ISAC.
Not every ISAC system fits in one box.

What an echo actually tells you



- **Delay** → range, $R = c\tau/2$
- **Doppler** → closing speed
- **Angle** → direction

Q: Which knob buys finer range, and which buys finer speed?

A: Bandwidth buys range resolution — a wider signal has a sharper peak in time. **Observation time** buys velocity resolution — you must watch long enough for the phase to drift.

Three questions for the rest of today

① **Why can't one signal be great at both jobs?**

The two jobs want opposite things. That gives us a curve of compromises, not a fix.

② **Which signal do we actually use?**

OFDM — the one Wi-Fi, 4G and 5G already transmit. No new radio.

③ **How do we share it between bits and echoes?**

The OFDM grid is a budget. Somebody has to decide who gets each cell.

Everything from here answers one of these three.
If you lose the thread, ask which one we are on.

They want opposite things (1/2)

	Sensing wants	Communication wants
Spectrum	Wide bandwidth, for resolution	Efficient bandwidth, for rate
Signal	Deterministic waveforms	Random, information-bearing symbols
Power	High transmit power on the target	Power efficiency per bit
Time	Continuous illumination	Low-latency bursts
Space	Wide or scanning beams	Narrow beams, one per user

Randomness is what carries information — and randomness ruins a clean echo.
That single sentence is the whole conflict.

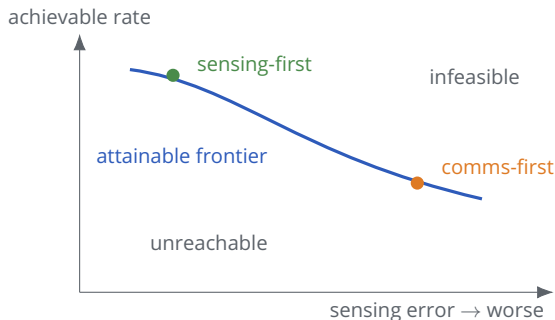
They want opposite things (2/2)

Sensing wants		Communication wants
Interference	A clean echo, no clutter	A clean constellation, no radar leakage
Domain	Fine steps in distance and speed	Fine slots in time and frequency
Hardware	Precise timing, high-power amplifiers	Spectral efficiency, many users
Latency	Low-latency tracking feedback	Tolerates delay off the critical path
Regulation	Frequency allocation rules	Emission and efficiency standards

Q: Is there a setting where both are optimal?

A: No. There is a **Pareto frontier**: past a point, every improvement in one is paid for out of the other. Design work is choosing where to sit on it, not escaping it.

The frontier, drawn once



Horizontal axis = sensing error: how far off the range and speed estimates are. **Lower is better**, so the good corner is up and to the left.

Every result in the rest of this lecture is a claim about one point on a curve like this.

Question 2: which signal do we actually use?

Question 1 is answered: the two jobs want opposite things, so there is a curve of compromises. Whatever signal we pick has to carry **both** jobs and sit somewhere on that curve.

The answer is OFDM (Orthogonal Frequency Division Multiplexing) — the signal Wi-Fi, 4G and 5G already transmit. No new radio.

Before we can read OFDM as a radar we need to know what it actually is. Most of you have only heard the name, so we build one from scratch, in four steps:

- 1 Bits become points.
- 2 Describing a point needs perpendicular axes.
- 3 The FFT is a set of perpendicular axes.
- 4 An OFDM symbol is one IFFT — and symbols make the grid.

Four steps, one example: by the end you have built the grid this lecture is about.

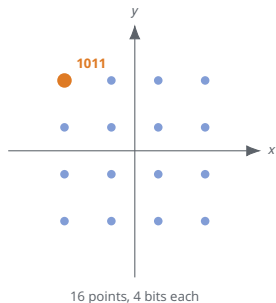
Step 1: bits become points

Say we want to send these bits:

1011 0010 1110 0001 ...

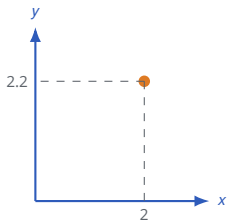
Cut them into groups of four. Each group picks **one of 16 points** on the map at right. That step is called **16-QAM**.

A point is just two numbers, (x, y) . So four bits became one point, and a long bit string became a **list of points**.

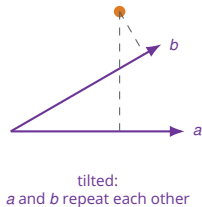


Bits $\rightarrow$ points. Nothing has been transmitted yet;
we have only chosen what to say.

Step 2: describing a point needs perpendicular axes



perpendicular:
two numbers, no overlap



In 2-D you need two axes, in 3-D three — and they must be **perpendicular**. Tilt them and each axis partly repeats what the other already said: the two numbers overlap, and reading one leaks into the other.

A signal is a point in a space with many dimensions.
To describe it cleanly we need many axes, all perpendicular to each other.

Step 3: the FFT is a set of perpendicular axes

Two dimensions. The 2-point FFT:

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a+b \\ a-b \end{pmatrix}$$

Row one: the **common part**. Row two: the **difference**. The rows are perpendicular — $1 \cdot 1 + 1 \cdot (-1) = 0$ — so the two answers do not overlap.

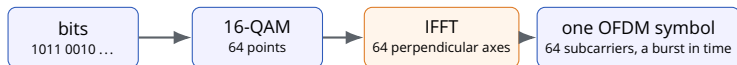
Four dimensions. Same idea; $j = \sqrt{-1}$ lets an axis rotate:

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{pmatrix}$$

Any two rows: multiply entry by entry (flip the sign of j in one of them), add — you get 0. **Every row is perpendicular to every other.** Each row is one frequency.

FFT asks a signal: how much of you lies along each frequency?
IFFT is the reverse: give it the amounts, it builds the signal.

Step 4: an OFDM symbol is just an IFFT



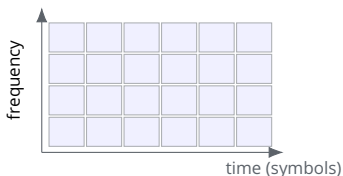
- Put one QAM point on each of 64 perpendicular axes.
- The IFFT adds them into one burst of time: a **symbol**.
- Each axis is a **subcarrier**, a narrow carrier of its own.
- Perpendicular $\Rightarrow$ no interference, even packed tight.

How tight? Spacing = $1/T$, one over the symbol length. The standard picks the number: 5G uses 15, 30, 60 or 120 kHz; Wi-Fi uses 312.5 kHz.

Send symbol after symbol and you get a **grid**: subcarriers down the side, symbols across. **One cell = one subcarrier for one symbol**.

That grid is the OFDM you already use every day.
Next slide: the same grid, read as a radar.

OFDM already carries a radar inside it



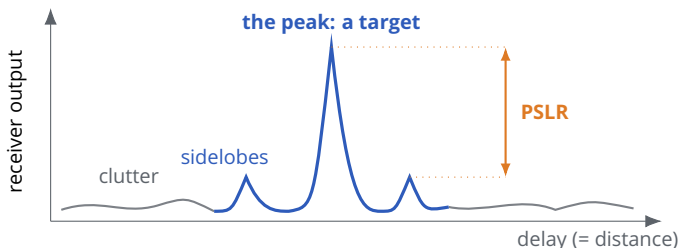
- Mature: Wi-Fi, 4G, 5G, 6G.
- The grid is already knobs.
- A cell serves bits or echoes.
- No new modulator needed.

Q: If OFDM is already good enough, what is left to design?

A: Which cells go to bits and which go to echoes. Every cell handed to sensing is a cell taken from data.

OFDM = Orthogonal Frequency Division Multiplexing

What a peak is, before we grade one



The receiver slides a copy of its own waveform across the echo. Where a target sits, everything lines up and adds — that spike is **the peak**. **Sidelobes** are ghosts of our own waveform; **clutter** is every echo we never asked for.

PSLR = how far the true peak stands above the tallest ghost.
That number is the sensing grade on the next slide.

Two scores, graded in two rooms

Communication score

Achievable data rate, summed over every RE assigned to communication.

Measured in **time-frequency**: put bits where the channel is good.

Sensing score

PSLR inside the region of interest — how far the true peak stands above the clutter.

Measured in **distance and speed**: make the peak stand tall where the targets actually are.

The same grid is graded twice,
by two examiners in two different domains.

Whoever picks first, wins

Communication-centric

- 1 Give the good REs to bits.
- 2 Then tune leftover powers.
- 3 Sensing takes the remainder.

Rate stays optimal; PS LR is merely good inside the RoI.

Sensing-centric

- 1 Lock the peak shape inside the RoI.
- 2 Then tune the rest of the grid.
- 3 Bits take the remainder.

PS LR is locally perfect; rate approaches, not reaches, optimal.

Q: Both designs claim “almost no loss”. How can that be?

A: Because the constraint is only enforced inside the **region of interest**. Narrow the room you must be perfect in, and the rest of the grid is free for the other function.

Three things to take home

- 1 **One waveform, two jobs.** The signal a base station already sends comes back as an echo, and that echo is measurement nobody had to upload.
- 2 **The two jobs want opposite things.** Sensing wants a wide, steady, predictable signal; data wants an efficient, bursty, random one. So there is a curve of compromises, not a fix.
- 3 **On OFDM, picking a point on that curve is a scheduling decision.** Which cells of the grid carry bits, which carry echoes, and who picks first.

ISAC is not a new radio.

It is a new question asked of the radio we already have: who gets each cell?

What is actually unfinished

Settled enough to build

- The tradeoff is understood.
- OFDM-based ISAC works.
- The RE grid can be allocated.

Still open

- A killer application.
- Security of a shared waveform.
- AI for waveform design.
- Space-air-ground reach.

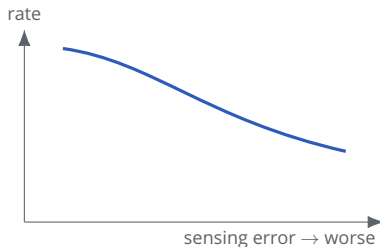
Q: Which of these is not an engineering problem?

A: The first. A waveform can be optimised; a market cannot. ISAC has a technology looking for the application that pays for it.

SAGIN = Space-Air-Ground Integrated Network

Before you go: pick a point on the curve

Two minutes, on paper, handed in on the way out. Same curve as before.



- 1 Mark where a **self-driving car** should sit. Mark where a **home Wi-Fi link** should sit.
- 2 Name one resource the two designs would allocate differently.
- 3 Answer in one sentence:
what does sensing cost the bits in your example?

Appendix: the acronym decoder ring

Area	Term	Meaning / why it matters
System	ISAC	Integrated Sensing and Communication
System	DFRC	Dual-Function Radar Communication; ISAC inside one radio
System	IMT-2030	The ITU framework naming ISAC as a 6G usage scenario
Waveform	OFDM	Orthogonal Frequency Division Multiplexing
Metric	PSLR	Peak-to-SideLobe Ratio; true peak above the clutter
Metric	SINR	Signal to Interference-plus-Noise Ratio
Resource	RE / RoI	Resource Element (one grid cell); Region of Interest (the distances and speeds we care about)

Appendix: the quantities behind every plot

Delay τ range, $R = c\tau/2$

Doppler ν speed, $v = \nu c/(2f_c)$

Bandwidth buys range resolution

Dwell time buys velocity resolution

Rate $\sum \log_2(1 + \text{SINR})$ over REs

Sensing error has a hard floor set by bandwidth, power and time

PSLR peak over worst sidelobe, in the RoI

Rate is a sum you maximise.
Sensing error has a floor you cannot go below.

Sensing accuracy is bounded from below.
That is why the frontier bends.

- **Zhu Han (Univ. of Houston), “Integrated Sensing and Communication for 6G”**

The invited talk this lecture is built from. Slides and code: wireless.egr.uh.edu.

- **ITU-R M.2160: IMT-2030 framework**

Where ISAC becomes an official 6G usage scenario.

- **Cross-domain dual-functional OFDM waveform design**

IEEE JSAC 2024. Source of the RE-allocation results.